Video Transcript - Part 14

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Hey everyone, welcome to part fourteen. In this video we're looking at AWS managed AI services, ready made services that add intelligence to your applications without you training a single model. We'll cover five of them: Amazon Comprehend, Polly, Transcribe, Rekognition, and Lex.

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Here's the plan. We'll start with what managed AI services are and why they're useful. Then we'll get an overview of all five, and go through each one in turn. After that we'll see how they can work together, and finish with when to use them instead of building your own model in SageMaker.

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So what are AWS managed AI services? They're pre-trained models that AWS runs for you, and you use them through simple API calls. You send in text, audio, or an image, and you get results back. You don't need machine learning expertise, there's no model to train or servers to manage, it scales automatically, and you pay only for what you use. They're ideal when your task is common, like understanding text, recognizing speech, or analyzing images.

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Here are our five services at a glance, organized by what goes in and what comes out. Comprehend takes text and gives back insights. Polly takes text and gives back spoken audio. Transcribe takes speech and gives back text. Rekognition takes images and video and gives back labels, faces, and more. And Lex handles conversations, powering chatbots and voice bots. A good way to remember them is text to insight, text to speech, speech to text, picture to labels, and conversation to bot.

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Let's start with Amazon Comprehend, AWS's natural language processing service. It reads text and finds meaning in it. Sentiment analysis tells you whether text is positive, negative, neutral, or mixed, which is great for reviews and support tickets. Entity recognition picks out people, places, organizations, and dates. It can also extract key phrases and detect the language. And PII detection finds personal information like names and addresses so you can redact it. If you need your own categories, custom classification lets you train it on your labeled examples.

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Next is Amazon Polly, which turns text into lifelike speech. It offers many voices across many languages, including neural voices that sound very natural. With SSML, a markup language for speech, you can control things like pauses, emphasis, speed, and pronunciation. Lexicons let you define how specific words are said, like product names or acronyms. People use Polly for e learning narration, accessibility features, voice in apps, and reading articles aloud.

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Amazon Transcribe goes the other way, converting speech into text. It works on recorded files in batch mode, and on live audio with streaming. It can identify different speakers in a conversation, and you can add a custom vocabulary so it recognizes domain specific terms and names. It can automatically identify the language being spoken, redact sensitive information, and generate subtitles. Typical uses are call center transcripts, meeting notes, and captioning video.

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Amazon Rekognition analyzes images and videos. It can detect objects and scenes, so it can tell you an image contains a dog, a beach, or a car. It does facial analysis, including detecting faces, comparing two faces, and searching for a face in a collection. It can read text that appears inside images, and content moderation flags inappropriate or unsafe content. If you need to spot your own objects, like your company's products, Custom Labels lets you train a model with your own images.

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Amazon Lex lets you build chatbots and voice bots, and it uses the same technology that powers Alexa. It has speech recognition and language understanding built in. Lex works with four key ideas. An intent is what the user wants, like booking a hotel. Utterances are the different phrases people might use to ask for it. Slots are the details Lex needs to collect, like the city and the dates. And fulfillment is the action that runs once it has everything, often an AWS Lambda function.

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These services are building blocks, and you can combine them. Picture a customer support system. Transcribe turns call recordings into text. Comprehend analyzes that text to find sentiment and the key issues. Lex runs a chatbot that handles common questions. And Polly speaks the answers out loud. Each step is just an API call, so you can mix and match them to fit your use case.

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So when do you use managed AI services, and when do you use SageMaker? If your task is a standard one, like sentiment, speech, or image labels, and you want results quickly without ML expertise, managed AI services are the right choice. If you need a custom model with full control over the data, algorithm, and training, that's where SageMaker comes in. And there's a middle ground. Options like Comprehend custom classification and Rekognition Custom Labels let you tailor a managed service with your own data.

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So to recap, managed AI services give you pre-trained intelligence through simple APIs. Comprehend understands text, Polly speaks it, Transcribe listens, Rekognition sees, and Lex holds the conversation. They're fast to adopt, and you reach for SageMaker when you need something fully custom. Thanks for watching, and I'll see you in the next one.